PHARMACOLOGY LAB I (2026-27)
PHARMACEUTICAL DOSAGE FORMS

SAYA MOHAMMED OMAR

MSC. PHARMACOLOGY

DOSAGE FORMS

- Dosage forms are **drug products** suitable for administration of a specific dose of a drug to a patient **by** a particular route of administration.



In general there are three types of dosage forms according to physical properties of the drug:

1. **Solid** dosage form:

-Tablet and capsule.

2. **Liquid** dosage form:

-Syrup and suspension.

3. **Semisolid** dosage form:

-Ointment and cream.

1.SOLID DOSAGE FORM:

Tablets and capsules:

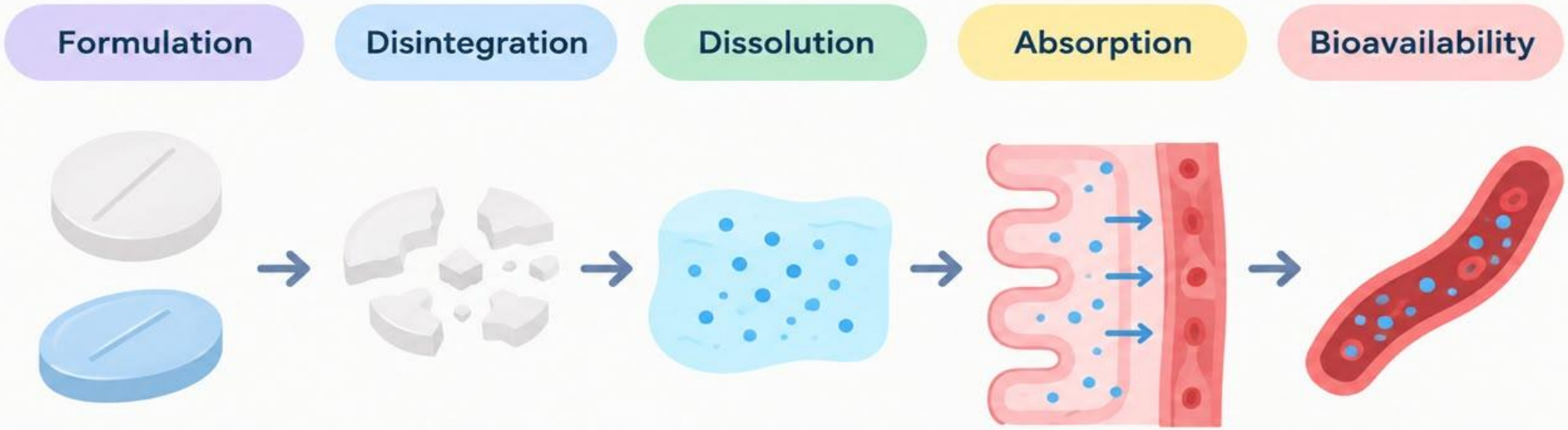
Tablets and capsules are the most common preparation for oral administration because:



1. They are suitable for mass production.
2. They are stable.
3. Convenient for use.
4. Can be formulated to release the dug immediately after ingestion or to release it over the period of hours.

A/TABLETS:

- Tablets are made by compressing a mixture of powdered drug and an inert ingredients into a relatively hard tablet.
- Tablets must be disintegrated after it is ingested, and then the drug must dissolve in gastrointestinal fluids before it can be absorbed into circulation.
- Different tablet formulations may give rise to differences in the oral bioavailability of drugs because of the variations in the rate and extent of tablets disintegration and drug dissolution.



Differences in formulation can change
the amount of drug that reaches the bloodstream
(bioavailability).

TYPES OF TABLETS:

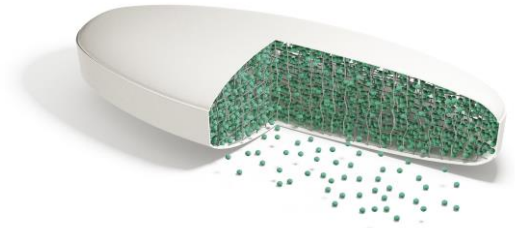
1- ordinary tablets.



2-Enteric coated tablets:

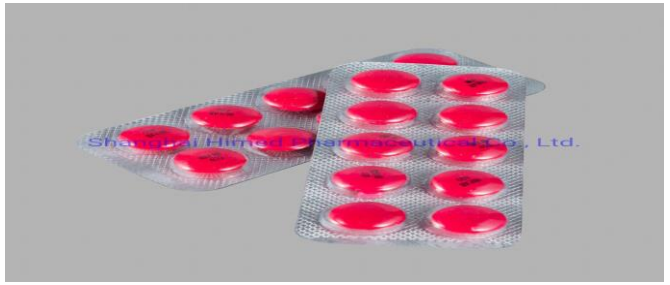
- Used to protect drugs that would be destroyed by gastric acid or the drug is an irritant to the gastric mucosa, The drug might be created with various types of coatings.

- **These enteric-coated tablets will not disintegrate in gastric acid but will be broken down in a more alkaline medium (small intestine).**



3-Sustained release tablets:

Designed to release their content over extended period of time, by controlling diffusion and controlled dissolution. It is composed of many coats or layers. Each coat or layer is disintegrated at a certain time to prolong the action of the drug. This type of dosage form is preferable for elderly and psychiatric patients.



4-Sugar-coated also called pills:

They are always spherical in shape. The sugar coating is to mask the unpleasant taste of the active substance, also to enlarge the size of the pill for very small quantities of drugs.



5-Lozenges: flavored tablets intended to dissolve slowly when held in the mouth, containing substances that are released to act in the mouth or throat.



6-Effervescent tablets: tablets that are dissolved in water before taken by the patient.



7-Chewable tablets: tablets that are chewed or crushed in the mouth of patient before swallowing it.

B/CAPSULES

Capsules consist of a cylindrical body .They are usually made of gelatin which is quickly dissolved in the GI fluid, there are two types of capsule:

1. Hard gelatin capsule enclosing drug as powder.



2. Soft gelatin capsule containing drug as liquid form.



Like tablets, there are sustained release capsules in which dissolution is controlled by means of inert polymers coating over small pellets of drug enclosed in a capsule; these pellets will gradually breakdown in the body fluids.



2-LIQUID DOSAGE FORMS:



Drug solutions and particle suspensions are the most common liquid pharmaceutical preparation and may be formulated for oral, parental, or other routes of administration.

These types of preparations provide a convenient method of administering drugs to pediatric and other patients who cannot easily swallow solid dosage.

A/Solution and Suspension for oral administration: are often sweetened and flavored to increase palatability.

1-Syrup: they are sweetened aqueous oral solution in which the sugar concentration is about 67%, to inhibit microorganism growth in the product, mostly used for drugs with unpleasant taste.

2-Suspensions: are powder suspended into aqueous solvent which must be shaken each time before taking the drug by the patients.

3-Elixirs: is a hydro alcoholic solution, in which the concentration of alcohol varies from 5-20%. Alcohol is included in elixirs as a solvent for drugs that are not sufficient soluble in water alone.


Syrup
Suspension
Elixir
%
Thick + sweet
Shake before use
Contains alcohol

B/Sterile solutions and suspension for injection

Are available for parenteral administration with a needle and a syringe or with an intravenous infusion device.

- Many drugs are formulated as sterile powders for reconstitution with sterile fluid at this time the drug must be injected, because the drug is not stable for long periods of time in solution and we have two types of injections:



I- Ampoules

2-Vials

- C/Sterile ophthalmic solution and suspensions: are suitable for administration with a dropper into the conjunctival sac, e.g. Eye drops.



- D/Solutions for oral hygiene (mouth washes and gargles).



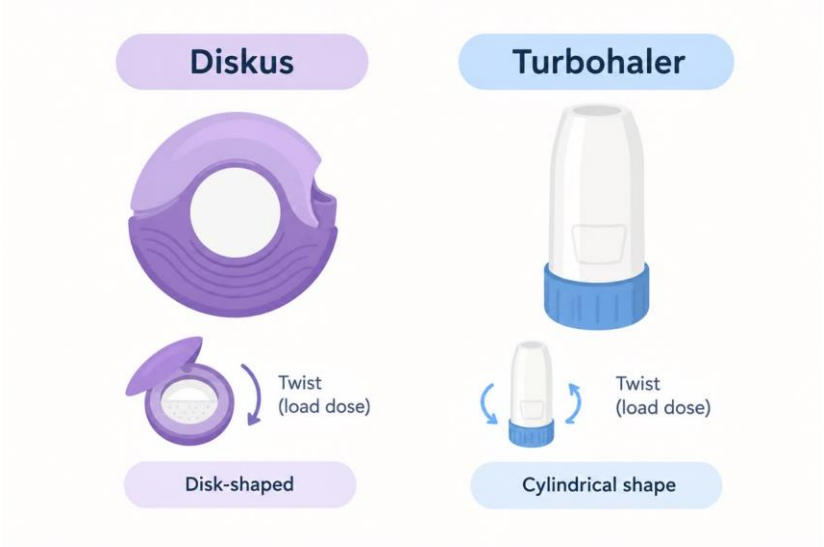
3- AEROSOLS:

- Are type of drug preparation administered by inhalation through the nose or mouth, they are useful for treating respiratory disease or disorders because they deliver the drug directly to the site of action and thereby minimize the risk of systemic side-effect.
- The devices that contain the drug dispersed in a pressurized gas and designed to deliver a dosage each time they are activated by the pump.

TYPES OF AEROSOLS:

1-Metered aerosols.





2-Powder aerosols. (diskus and turbuhaler)

3-Nasal spray



4-SEMISOLID DOSAGE FORM OINTMENT, CREAM, GELS AND LOTIONS:



Ointments and creams are semisolid preparation intended for topical application of a drug to the skin or mucous membrane. These products contain the active drug that is incorporated into a vehicle which enables the drug to adhere to the tissue for a sufficient time to exert its effect.

- **Ointment**: oil base
- **Cream:** water soluble base
- **Gels:** colloidal suspension of hydrated base
- **Lotions:** are liquid preparations that are often formulated as oil-in-water suspension used for treating dermatological disorders.

Ointment	Cream	Gel	Lotion
			
			
Greasy, oily base	Soft, emollient base	Water-based, non-greasy	Thin, liquid base



5-Suppositories:

are products of a conical shape, the base is either wax or hard paraffin which will melt at the body temperature.

Suppositories are usually intended for rectal or vaginal administration and may provide either localized or systemic effect.

6-Powders and Granules: bulk powder containing the potent drug mixed with inert diluents, the doses either folded separately in paper, sachets or packed in one bottle.



7-Skin patches:

They are drug preparations in which the drug is slowly releases from the patch for absorption through the skin into the circulation. They act either locally or systemically.

Such dosage forms are most suitable for drugs that are effective at a relatively low dosage and have sufficient lipid solubility to enable the skin penetrations.



- **Enemas**



- **Gum**



- **Film**



- **Plasters**
- **Soap**
- **Shampoo**
- **Pastes**









- **Dressing**



- **Douche**

